

The Basin of Mexico Hydrogeological Database (BMHDB): Implementation, queries and interaction with open source software

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ABSTRACT

Integrated Water Management at the Basin level is a concept that was introduced in the 1990s and is a goal in every national and local water management plan. Unfortunately this goal has not been achieved mainly due to a lack of both tools and data management, as data must be gathered from different sources and in different formats. Compounding this problem is the fact that in some regions different water agencies are in charge of water supply as is the case in the Basin of Mexico, in which Mexico City and its Metropolitan Zone are located. The inhabitants of the Basin of Mexico, which comprises five different political entities and in which different agencies are in charge of water supply rely on the Basin's aquifer system as its main water supply source. However, a regional hydrogeological database in this area does not exist which is why the use of both a Relational Database Management System (RDBMS) and a Geographic Information System (GIS) is proposed in order to improve regional data management in the study area. Data stored in this new database, the Basin of Mexico Hydrogeological Database (BMHDB) comprise data on climatological, borehole and run-off variables, readily providing information for the development of hydrogeological models. A simple example is used to show how geostatistical analysis can be done using the data directly from the BMHDB. The structure of the BMHDB allows easy maintenance and updating, representing a valuable tool for the development of regional studies.

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Software availability¹

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Required software (freely available)

GRASS: grass.itc.it

PostgreSQL: www.postgresql.org

PostGIS: www.refractory.org/postgis

R statistical software: www.r-project.org

Platform: platform independent (Linux, Windows, OsX)

1. Introduction

The concept of water management evolved from a piecemeal approach to that of a Basin level management after the International Conference on Water and Environment in 1992, which

resulted in the Dublin statement (ICWE, 1992). This concept shifted to Integrated Water Resources Development and Management in Rio, later in that same year as expressed in chapter 18 of Agenda 21 (UNCED, 1992). Behind this idea was the introduction of both land and water related aspects of water management at the Basin level as well as stakeholder participation. However, this seldom occurs in practice, mainly due to a lack of both adequate data management and proper tools to achieve an integrated river Basin approach.

In an Integrated Water Management approach, the development of regional hydrogeologic models is required in order to predict the impact of different land and water management policies in the future and all data required should be easily accessible to decision makers and modelers. Data accessibility to several users can be provided by a Relational Database Management System (RDBMS) and it can incorporate data such as location of wells, pumping rates, groundwater table elevation, lithology records, concentration of trace metals as well as chemical and physical parameters of the groundwater. The same database can include climatological variables (e.g. precipitation, temperature, evaporation) thus making it possible to undertake spatial and time series analysis of these variables.

The use of a Geographic Information System (GIS) can help to visualize and update the existing database as superposition of different thematic layers can be accomplished in order to verify

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existing data and existing modules can be used on the database (e.g. geostatistics). The database structured in this way can be queried with the use of Structured Query Language (SQL) statements and new tables can be formed from existing data.

Although the use of a GIS can help to visualize existing data through the simultaneous display of different layers or 3D views, statistical analysis can be done without the use of a GIS. The relational database PostgreSQL was selected in the present work as it can be linked to the statistical language R (R Development Core Team, 2005) which is an open source project similar to the S language. R provides libraries for statistical analysis. Another advantage of PostgreSQL is that it can be linked to the open source GIS GRASS which provides tools for raster, vector and point analysis as well as tools for image processing. In addition, PostgreSQL can handle spatial attributes such as points, polygons or lines by using the postGIS extension which makes it possible to undertake spatial queries to the database.

The objective of this work is to present a database management system for the Basin of Mexico, providing a framework that can be applied to any other study area. The database comprises data previously available in different locations and formats in order to make it available to all interested users. This database can be accessed using open source software, freely available from the Internet in order to make data accessible to people who cannot acquire proprietary software due to its cost.

2. The Basin of Mexico

The Basin of Mexico (referred to as Basin in the remainder part of the paper) with an approximate area of 9600 km² encloses one of the largest cities in the world: Mexico City and its Metropolitan Zone (MCMZ). The Basin is located in the central part of Mexico and

is enclosed by mountains as high as 5500 masl (Fig. 1) while the valley's mean elevation, where Mexico City is located is near 2240 masl.

The Basin's aquifer system is the most important part of the water supply system to its inhabitants and its exploitation started in 1847, when the first well was drilled (Ortega and Farvolden, 1989) a number which by 1990 had increased to 3537 officially registered wells in the MCMZ (NRC, 1995). This heavy dependence on the aquifers has had its toll and a decline in the potentiometric level of up to 80 m was recorded by 2002 in some areas (Edmunds et al., 2002). Compounding this problem, the Basin comprises five different political entities (Fig. 1); accordingly, different governmental agencies are in charge of water supply, the most important being the *Comisión Nacional del Agua* (CNA) and the *Dirección General de Construcción y Operación Hidráulica* (DGCOH). The CNA has under its charge the *Gerencia Regional de Aguas del Valle de México* (GRAVAMEX) which in conjunction with the DGCOH operates the water supply infrastructure for the MCMZ. However, water management at the Basin level is not fulfilled as these agencies operate on their own, making it difficult to share information between them.

The need for accessible and up-to-date data at the Basin level is shown in recent studies undertaken within the Basin of Mexico as they have considered only subareas of the Basin or rely on short-term records such as Birkle et al. (1998), who used rainfall data for the 1980–1985 period to develop a “long-term” water balance.

3. Improving data management

Data required for any type of surface or groundwater study in the Basin are currently spread throughout different agencies in charge of water supply and even within these agencies data are

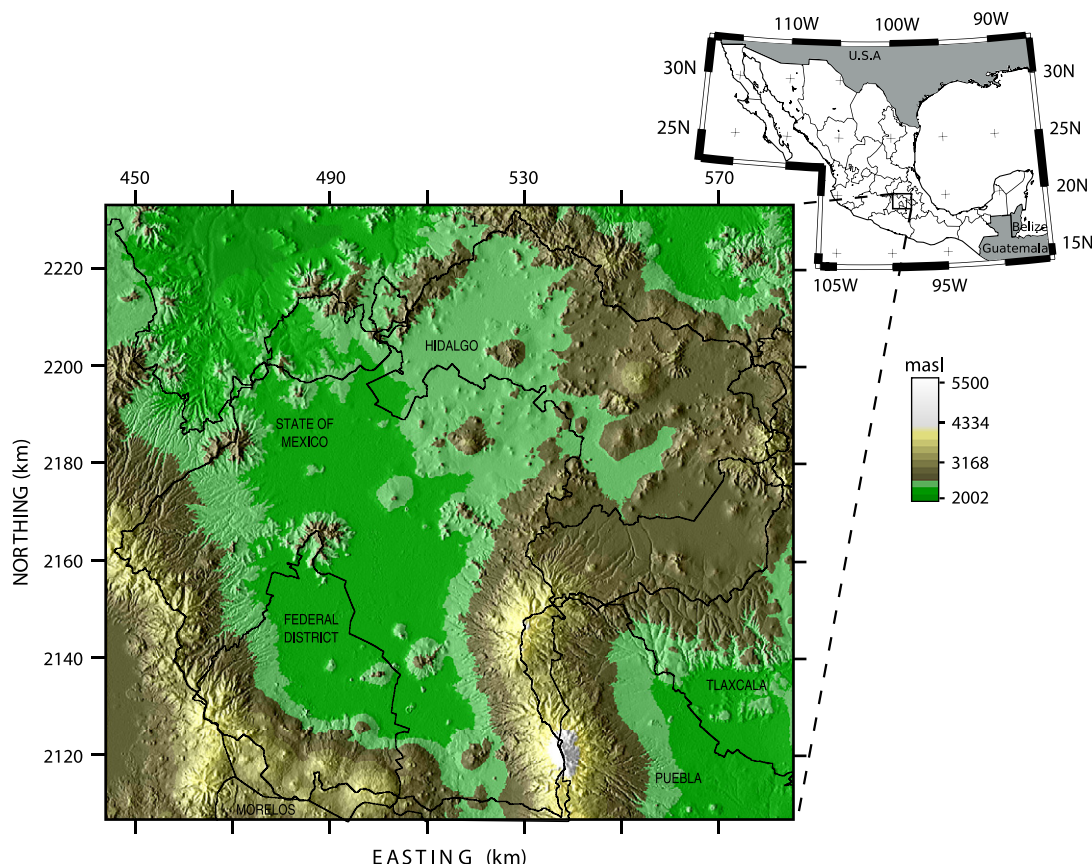


Fig. 1. Location, Digital Elevation Model and political boundaries for the Basin of Mexico. Coordinates are in kilometers for UTM-14; elevations are in meters above sea level.

found in different reports. Furthermore, the existing databases are limited to particular data such as climatological or run-off data. In order to improve water management in the Basin and to foster an Integrated Water Management approach in the study area, the Basin of Mexico Hydrogeological Database (BMHDB) has been developed using both a Relational Database Management System (RDBMS) and a Geographic Information System (GIS). Regarding well-related data, the BMHDB comprises monitoring wells from both CNA and DGCOH, allowed extraction volumes from those wells registered at the *Registro Público de Derechos de Agua* (REPGA) and annual extraction rates for those wells registered at DGCOH. Additionally, the database contains lithology records, pumping tests and chemical data for some of the wells. Climatological data (i.e. rainfall, minimum and maximum temperature) are available on a daily, monthly and annual basis, as are run-off data (volumes and flows). This information, which is currently distributed in the water supply agencies (DGCOH, CNA) and in previous studies realized in other areas of the Basin was gathered in order to develop this new database. The information gathered so far was obtained in different formats, such as spreadsheet files, shape files, hard-copy maps (e.g. soils and land-use), hand written tables and reports, which had to be processed and georeferenced in order to provide readily accessible data. The tables can be updated through the use of simple SQL command such as `insert` or `copy`.

3.1. Existing databases

There are currently two databases in Mexico which contain data required in hydrogeological studies. The databases that are available to any user are the *Extractor Rápido de Información Meteorológica* (ERIC), (IMTA, 1990) and *Banco Nacional de Datos de Aguas Superficiales* (BANDAS), (IMTA, 1995) which are briefly explained below.

3.1.1. ERIC

This database is distributed on one CD which includes nation-wide daily meteorological data: rainfall, pan evaporation, minimum temperature, maximum temperature, average temperature, storm (0 = no storm, 1 = storm), overcast conditions (0, 1, 2) and hail. The data stored on the CD have to be copied to the user's hard disk and accessed through the DOS command line. To query this database the user's input is required; the user needs to type the desired query in a specific order on the command line: (1) variable selection (e.g. rainfall, evaporation), (2) station selection (one station, all stations, rectangular, polygon or state-wide selection) and (3) time interval (one day, one period or one period over several years). The output of this query is an ASCII file with text and data which needs to be formatted and cleaned in order to undertake any type of statistical analysis.

3.1.2. BANDAS

This database was developed by the *Instituto Mexicano de Tecnología del Agua* and as ERIC, it comprises nation-wide data for Mexico. It is distributed as six CDs which are available from IMTA; the first of these CDs provides the installation program and is required in order to access the data. The information stored in this database is organized in 13 hydrological regions and in order to query it the user has to make a predetermined number of selections which can only be made through a scroll-menu. First the user has to select the hydrometric station of interest and then click on an icon to query the selected station which brings up another window. On this window the user is presented with different options through selection boxes which cannot be selected simultaneously (and which in some cases are repetitive): monthly data, mean daily flows, flow records greater than a user-defined threshold, average and extreme annual flows, daily hydrometric data (flow and

volume), monthly and annual hydrometric data. The way in which this database is structured makes it tedious to gather data for more than one year. As ERIC, the output file contains text and relevant data.

3.2. Drawbacks of the existing databases

In order to undertake any type of statistical analysis with the output data from either ERIC or BANDAS, the data have to be processed in order to clean from them additional information printed by these databases. In summary, the existing databases for climatological and stream-flow data have the following drawbacks.

- They need to be installed on computers running proprietary software, which means that they are not platform independent thus hindering their access.
- The output of these databases has to be processed in order to be analyzed as it contains text (e.g. NA or sentences) within the data.
- The data stored on these databases comprise only a fraction of the required input in any type of hydrogeological study.

4. The Basin of Mexico Hydrogeological Database (BMHDB)

As previously explained, hydrogeological information is spread throughout different agencies in the Basin of Mexico. In order to improve Water Management in the Basin it is first suggested to improve data management through a central database system which provides remote access in order to facilitate its updating and maintenance. The development of the Basin of Mexico Hydrogeological Database (BMHDB) comprised three main procedures as illustrated in Fig. 2 and which consisted of:

1. Data transcription: this stage consisted of transcribing the data acquired as hard-copy reports such as location of wells, lithology records and groundwater table elevations.

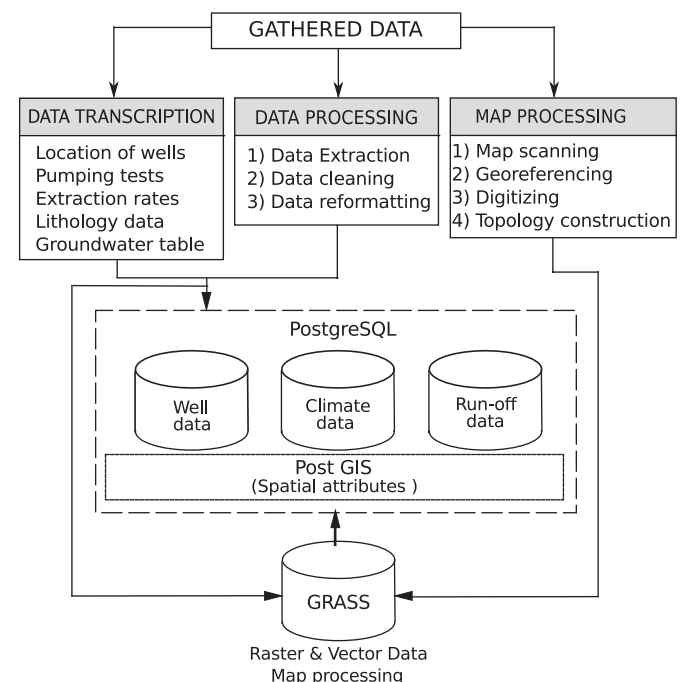


Fig. 2. Development of the Basin of Mexico Hydrogeological Database and interaction of its different components.

2. Data processing: data from the existing databases or data provided in spreadsheet formats were extracted and reformatted in a format usable by PostgreSQL. Spatial properties were reformatted according to the requirements of postGIS.
3. Map processing: hard-copy maps (e.g. geology, land cover and edaphology) had to be digitized and georeferenced before being processed. The processed maps are stored as both vector and raster maps in the GRASS database.

4.1. Data description

The structure of the BMHDB currently comprises 13 different tables as illustrated in Fig. 3 which can be divided in three sub-databases: climatological records, well records and run-off data. Some tables are organized by the agency which has the data (e.g. CNA or DGCOH) in order to facilitate the task of updating the database as it avoids duplication, however, these records could be merged into one table. The relevant fields of each table are shown in Fig. 3; as the BMHDB is a relational database, all tables are related by the *id* field of each well (Fig. 3a), climatological station (Fig. 3b) or gauging station (Fig. 3c).

The BMHDB comprises data at the Basin scale; this was accomplished by gathering data from different governmental agencies; it should be stressed that no attempt has been done to date in order to integrate all these data. As illustrated in Fig. 3 different tables are related to each other, and all data have a spatial reference stored in the *coords* field, which avoids confusion when dealing with different coordinate systems. This new database also integrates climatological and hydrometric information which were retrieved from BANDAS and ERIC (Fig. 4), improving the way in which this information can be retrieved and visualized. The BMHDB also comprises spatially variable information in both

vector and raster format which can be used in distributed hydrogeological modeling such as to analyze the impact of urban growth on aquifer recharge. The development of such analysis requires land cover map for different years, as illustrated in Fig. 5 which shows such a map for both 1978 and 1985, the spatial distribution of geological and soil units in the Basin, and the spatial distribution of rainfall using its correlation with topography, as done by Carrera-Hernández and Gaskin (2007b) who used local Kriging with External Drift to develop rainfall maps and Kriging with External Drift for both minimum and maximum temperature in the study area.

When developing the BMHDB, some data were missing from the original sources such as the elevation of each well; to complete the *wellsdgcoh* or *wellscna* tables with the *elev* field, the DEM was queried for those wells which did not have this information as explained in a later section. In addition, in order to ease monthly and annual statistical analyses, tables monthly and annual were developed from daily data by using only those stations with complete records. It is worth mentioning that data stored in the BMHDB comprises officially registered wells, however, nonregistered wells exist throughout the Basin. This adds another uncertainty factor to be considered when groundwater flow modeling is undertaken, however, their impact may be negligible, as the largest drawdown rates in the Basin are caused by wells used for municipal water supply (Carrera-Hernández and Gaskin, 2007a).

4.2. Socioeconomic data

Sustainable water management also comprises both social and economic aspects, therefore the database can also be extended to include these types of data. Socioeconomic data are available on a municipal basis, thus a subdatabase with a main table called *municipalities* (which is comprised of polygon data) can be linked to the tables described in the previous section through the *munid* field. Data stored in the municipal subdatabase comprise information on population, household income, number of rooms per household, municipal services (e.g. piped water, sewage, electricity) and crops grown (e.g. type and yield).

5. Querying the database

The BMHDB can be queried by using SQL statements for which knowledge of the database structure is required (Fig. 3). The information provided in this section aims to illustrating how the database can be queried and by no means aims to provide a review of SQL statements; interested readers are encouraged to read the PostgreSQL and postGIS documentation in order to undertake more complex queries. Generally speaking, the procedure to analyze the data stored in the database can be summarized in three steps: (1) the database must be queried and a new table is written with the data of interest (this can be done either within GRASS, R or the *psql* command line) using SQL statements, (2) the new table is used as input for the interpolation procedure and (3) the resulting map is saved as a raster file. An example is developed in the next section in order to illustrate the procedure.

In order to build the BMHDB, data stored in it were queried to obtain the elevation of wells or hydrometric stations. Borehole information compiled from the different water agencies in charge of water supply in the Basin comprised data on depth to the water table while each geologic stratum had an associated depth and not its elevation which is required in order to characterize the aquifer. To obtain the elevation of each borehole the DEM was queried using each borehole's coordinates in order to get their elevation which was stored as a table with two fields: *id* and *elevation*. The elevation column was added to the well's main table as column *elevation* (Fig. 3a). A new table was created in order to account

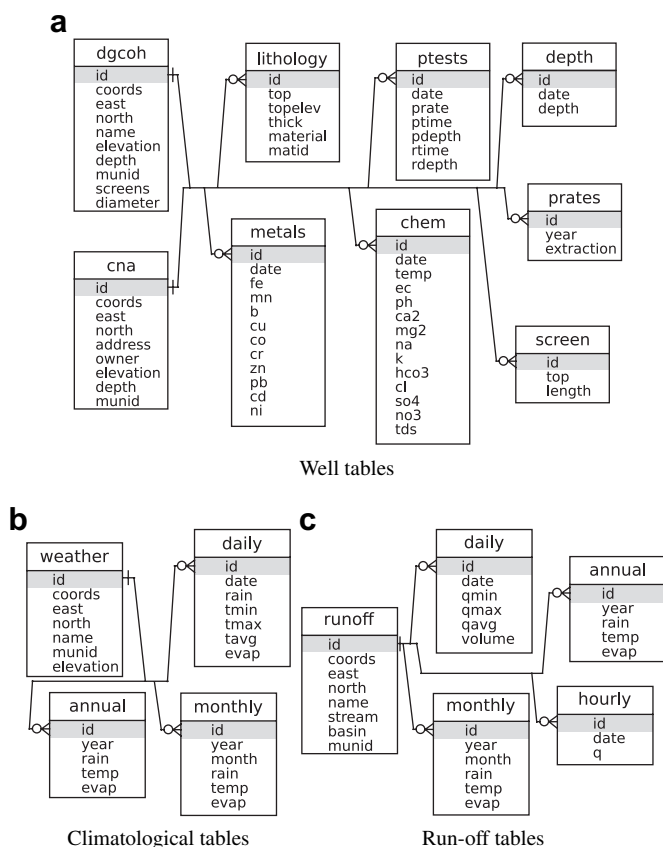


Fig. 3. Structure of the Basin of Mexico Hydrogeological Database.

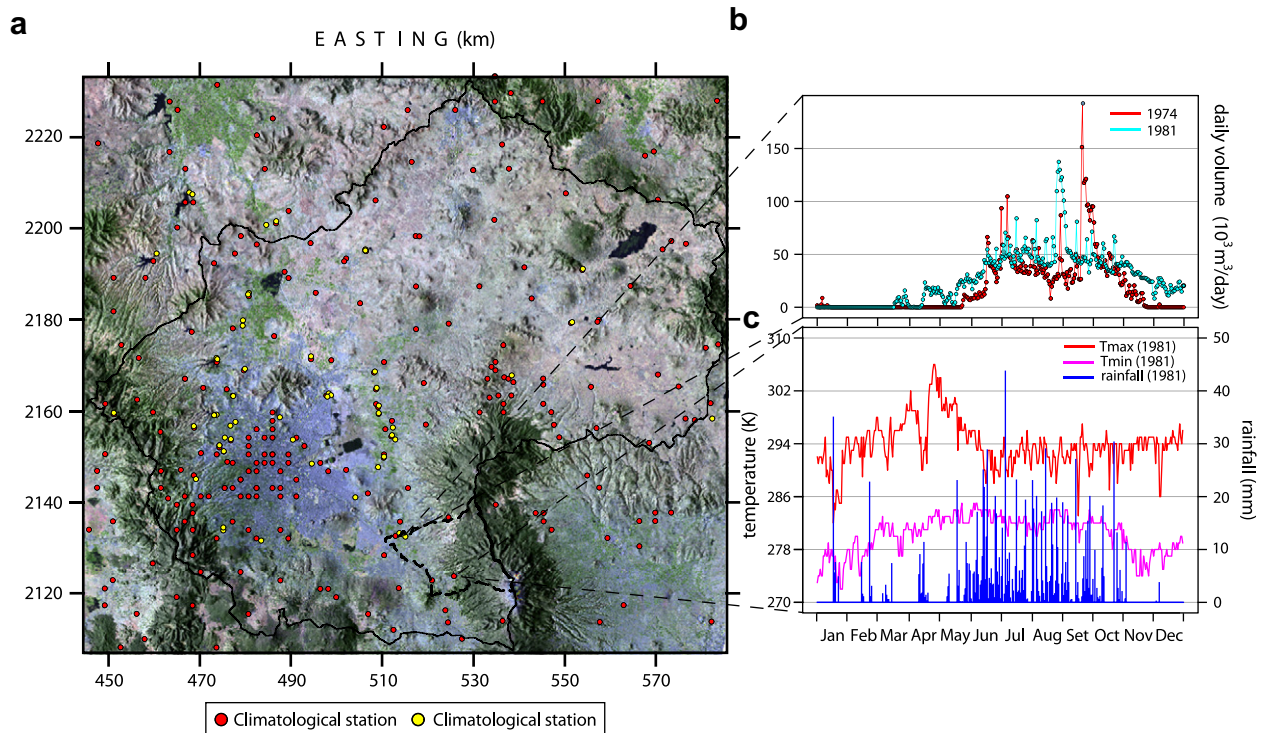


Fig. 4. Surface water related data: (a) spatial distribution of climatological and hydrometric stations, (b) daily river flow volume and (c) daily climatological data. The hydrometric and climatological stations are shown on a false color composite derived from LANDSAT-ETM+ imagery for March, 2000. Coordinates are in UTM, zone 14. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

for the elevation of each stratum by using the following SQL command:

```
CREATE TABLE lithodgcoh AS SELECT l.id, l.top
(d.elev-l.top) AS topelev, l.thick, l.material,
l.matid
```

```
FROM lithology AS l, dgcoh AS d
WHERE l.id=d.id;
```

The above command creates a table named lithodgcoh with fields id, top, topelev, thick, material and matid. The field topelev is computed by subtracting the top depth of each stratum

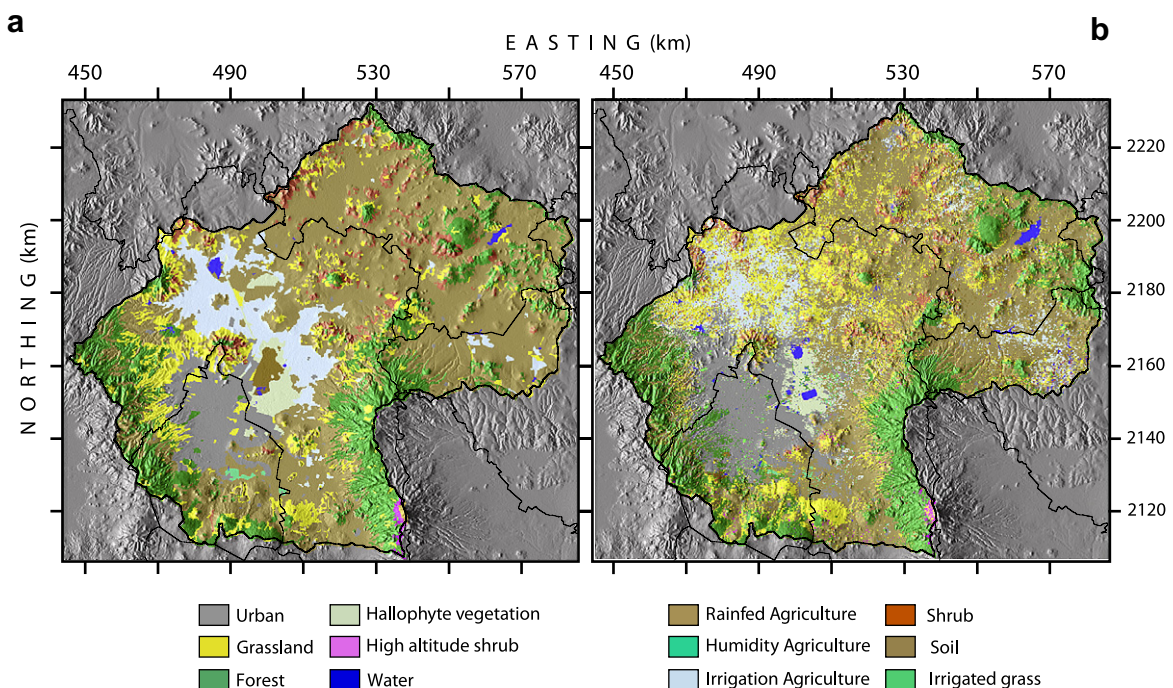


Fig. 5. Land cover map for the Basin of Mexico for two different years: (a) 1978 and (b) 1985. Coordinates are in UTM zone 14.

to the DEM elevation at the corresponding well (accomplished by the WHERE condition of the SQL statement).

5.1. Spatial queries

The BMHDB is a spatial database which means that data have spatial attributes such as coordinates (e.g. x–y, lat–lon), spatial reference (e.g. UTM zone) and datum (e.g. NAD-27). This information is stored as geometry by postGIS and allows queries involving spatial information. In order to include the `munid` field in the `dgcoh` table, the following spatial query was used:

```
SELECT w.id, m.id
FROM pai as w, municipalities as m
WHERE w.coords && m.coords
AND contains (m.coords,w.coords) and m.id<30000
ORDER by w.id;
```

This query selects the well and municipality `id` fields for those wells located inside a municipality whose `id` is less than 30 000 and evidently, other spatial queries can be done. Let us assume that a user wants to analyze the lithology records of those wells located in the Quaternary lacustrine deposits (Qla, cat 1); this selection is accomplished by using the postGIS function `contains`, the surface geology vector map and the well database as follows:

```
SELECT w.id FROM dgcoh as w, geology as g
WHERE w.coords && g.coords
AND contains (g.coords,w.coords) AND g.cat=1
ORDER by w.id;
```

Through the previous SQL command, those wells that are enclosed by polygons of category 1 (where category 1 = Qal) from table `dgcoh` are selected. The output is ordered by the well's field `id`. In this manner only those wells located inside one or more polygons can be found.

6. Interaction with other open source software

The reason why the BMHDB was developed was to provide readily accessible data for hydrogeological studies, which can be done by using different open source software, such as the R

statistical software (R Development Core Team, 2005), which provides tools for classical statistical tests, time series analysis and spatial interpolation, as different libraries can be used to undertake these tasks. One of these libraries is the GSTAT library (Pebesma, 2004) which can be used for spatial interpolation through different Kriging methods (i.e. Ordinary, Universal or local Kriging). Once the spatial distribution of a variable is obtained (i.e. temperature or rainfall), it can be stored as a GRASS raster map through the use of R's grass library (Bivand, 2000). This approach was used by Carrera-Hernández and Gaskin (2007b) to undertake a daily analysis of rainfall and both minimum and maximum temperature in the study area using a raster map (DEM) and time series data of these three climatological variables. The importance of the approach taken to develop the BMHDM is also illustrated by Carrera-Hernández and Gaskin (2007a) to undertake the first Basin-wide analysis of the evolution of the potentiometric level in the study area using satellite imagery along with time series data of pumping rates and groundwater table elevation. In order to show how these analyses can be undertaken, the current section provides simple examples on how data stored on the BMHDB can be readily accessed and analyzed using annual rainfall for 1979.

6.1. Geostatistical analysis of data

The methodology described is exemplified by (1) analyzing the correlation between annual accumulated rainfall and elevation for years 1972–1985 by developing a scattergram showing the correlation value between these two variables and (2) by developing a spatial map of accumulated rainfall for the Basin of Mexico in 1979 through the use of Kriging with External Drift (KED), with elevation as a secondary variable. Although different GUIs are available for R, its main advantage is that it can be used from the command line, providing flexibility and the capability of using scripts and accessing it in batch mode, allowing to undertake large amounts of statistical analysis. The commands required to access the database and analyze the correlation for the previously mentioned period are shown on Fig. 6 for which a brief description is given in order to illustrate the capabilities of the BMHDB. The goal is not to develop a brief tutorial and interested users are referred to the R project web page, which provides a listing of all the available packages and their documentation.

```
1 library(RPgsQL)
2 db.connect(host=NULL,port=NULL,dbname='BMHDB')
3 db.execute("SELECT a.id,a.year,a.rain,w.elevation
,w.east,w.north FROM annual as a, weather as w
WHERE a.year>=1972 AND a.year<=1983 AND a.id=
w.id AND w.elevation is not null AND a.rain is
not null",clear=F)
4 id<-db.read.column("id",as.is=F)
5 rain<-db.read.column("rain",as.is=F)
6 year<-db.read.column("year",as.is=F)
7 s1<-db.read.column("east",as.is=F)
8 s2<-db.read.column("north",as.is=F)
9 dem200<-db.read.column("elevation",as.is=F)
10 annualrain<-data.frame(id,year,east,north,dem200,
rain)
11 xyplot(rain~dem200|year,data=annualrain,ylab="
accumulated rainfall (mm)",xlab="elevation (
masl)",panel = function(x,y){panel.xyplot(x,y,
pch="+")+panel.abline(lm(y~x));grid.text(round
(cor(x,y),2),x=unit(1,"mm"),y=unit(1,"npc") -
unit(1,"mm"),just=c("left","top"))},layout=c
(4,3),ylim=0:2000)
```

Fig. 6. Analysis of annual accumulated rainfall in the Basin of Mexico using the R statistical language.

In order to be able to read and write to the GRASS database, R needs to be called from within GRASS from the command line, after which the required libraries must be loaded (Fig. 6, line 1): `spgrass6` is used to write/read data from GRASS, `RpgreSQL` is used to access PostgreSQL and `GSTAT` is used for the spatial interpolation. In addition, R automatically loads other libraries such as `grid` and `lattice` which were used to plot the different correlation values for 1972–1983 in Fig. 7. Once the libraries have been loaded, the database is accessed from R (Fig. 6, line 2) which in this case is being accessed on a local computer and so both the host and port options are set to null values. The database is queried using standard SQL commands (Fig. 6, line 3) and a dataframe is created in order to ease the statistical analysis which can be done with R (Fig. 6, lines 4–10). The scattergram plot showing both the correlation line and value (Fig. 7) were computed with line 11 of Fig. 6.

As previously mentioned, Kriging with External Drift is used in order to develop the spatial pattern of rainfall for 1979 which was chosen as it is the year that exhibits the largest correlation value between rainfall and elevation (Fig. 7). The spatial interpolation is undertaken through the commands shown in Fig. 8 in which lines 1–3 are used to load the required libraries. The Digital Elevation Model (Fig. 1), which is stored in GRASS is used as an auxiliary variable in the use of KED and is read in line 4. Using the annual data retrieved in the previous step (Fig. 7), a new dataframe for the selected year is created to facilitate the example (Fig. 8, line 8). A semivariogram is computed, visualized and used to undertake the spatial interpolation (Fig. 8, lines 9–14) and plotted using R's `sp` library (Fig. 8, lines 15–17) as shown in Fig. 9. The interpolated surface can be written to the GRASS database (Fig. 8, line 18) in order to be used in further studies such as a spatially distributed water balance.

7. Discussion

New tools are required to improve water management at the Basin level and to achieve a participatory approach. The BMHDB

provides both local and remote access as it uses the RDBMS PostgreSQL and can be queried with or without the use of a GIS as shown in this work. In addition, the way in which the database is structured avoids the need for data processing in order to undertake statistical analysis. In order to undertake any analysis on the data stored in the BMHDB its structure needs to be transparent to the end users.

The proposed database facilitates the task of compiling information as it can be easily updated and accessed due to the RDBMS client-server capabilities. Some information such as the wells' coordinates was verified with existing maps when available or by locating them in a map and checking if their location corresponded to that stored in the database.

The BMHDB provides data in spatial format thus allowing queries of spatial nature such as distance or polygon inclusion. Furthermore, the database can be processed with GRASS' raster and vector modules which allows the development of new maps by using map algebra for raster maps or unions and intersections for vector data. These modules can be used when developing groundwater flow models to calculate, for example, the width and length of a river reach within a finite difference model cell. Another advantage of the BMHDB is that its data can be used "as-is" to develop groundwater flow models through the use of the `r.gmtg` tool (Carrera-Hernández and Gaskin, 2006) which links the finite difference groundwater flow model MODFLOW with the GRASS GIS.

The development of the BMHDB represents a step towards Integrated Water Management in the Basin, as its data can be used to analyze the relationship between land cover change and groundwater; in addition, it also provides a framework to analyze water demand, as socioeconomic data can be also added to the database. Improving water management involves proper data management, a task that can be achieved using the framework presented in this paper as: (1) all tools used in the development of the BMHDB are open source software, which can be downloaded from the Internet (grass.itc.it, www.r-project.org, www.postgresql.org and postgis.refrains.net) and (2) these tools are cross-platform, meaning

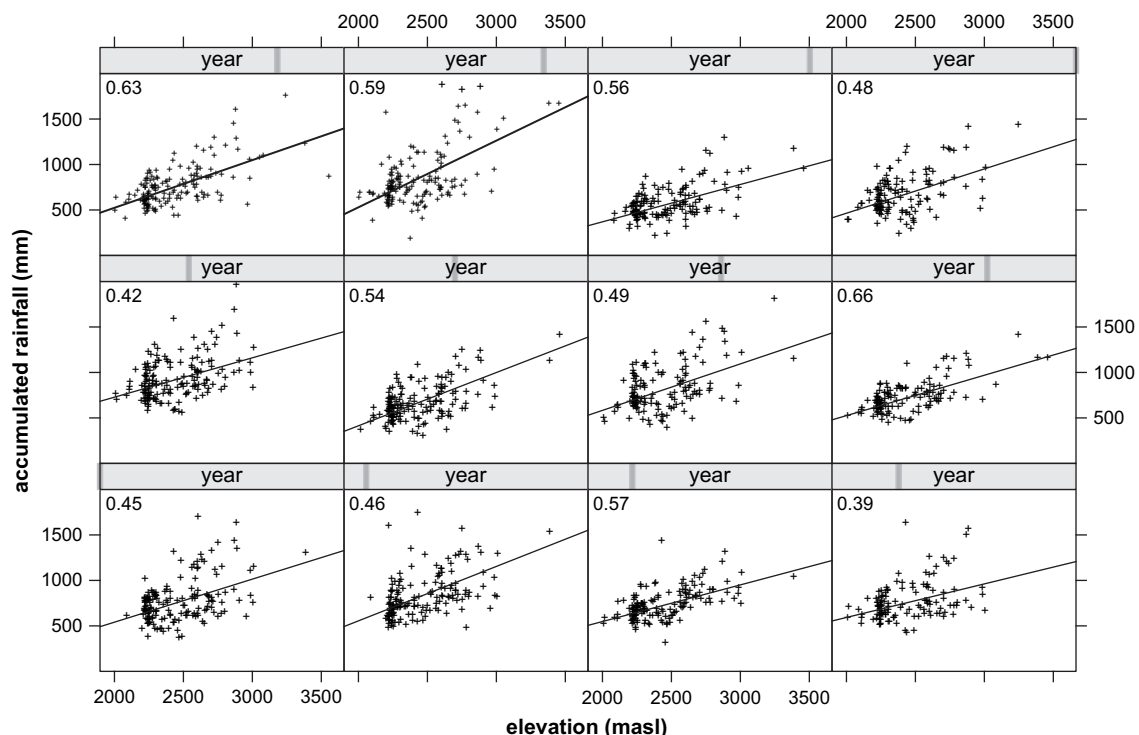


Fig. 7. Correlation between annual accumulated rainfall and elevation for years 1972–1983.

```

1  library(spgrass6)
2  library(gstat)
3  library(RColorBrewer)
4  elevation<-readCELL6sp("dem200")
5  ccacoords<-coordinates(elevation)
6  dem200<-elevation$dem200
7  ccagrid<-data.frame(ccacoords,dem200)
8  rain79<-data.frame(annualrain[annualrain$year
  ==1979,])
9  vgm79<-variogram(rain~dem200,~s1+s2,rain79,cutoff
  =40000)
10 sil<-max(vgm79$gamma)
11 nug<-min(vgm79$gamma)
12 vgmfit79<-fit.variogram(vgm79,vgm(nug,"Exp",15000,sil))
13 plot(vgm79,main="1979",model=vgmfit79)
14 rainked79<-krige(rain~dem200,locations=~s1+s2,
  data=rain79,model=vgmfit79,newdata=ccagrid)
15 coordinates(rainked79)=~s1+s2
16 gridded(rainked79)=TRUE
17 spplot(rainked79["var1.pred"],sp.layout=list("sp.
  points",stations,pch=19,cex=0.45,col="black"),
  pretty=TRUE,cuts=9,col.regions=brewer.pal(9,"
  Blues"),xlab="EASTING",ylab="NORTHING",scales
  =list(draw=TRUE))
18 writeRast6(rainked79,"kedrain79")

```

Fig. 8. Geostatistical analysis of annual rainfall in the Basin of Mexico for 1979 using elevation as a secondary variable.

that they can be used on different operating systems without restrictions. Data stored in the BMHDB can be accessed only by authorized users.

The importance of developing a reliable database for hydrological modeling has triggered the development of similar, larger and well-funded projects such as the Consortium of Universities for the Advancement of Hydrologic Science (CUAHSI) Hydrologic Information System. The framework used to develop the BMHDB can be extended to a national level in Mexico, in order to integrate different databases required to develop complex distributed numerical models. Evidently, the database would be greatly improved by integrating real time data, for which the participation of Mexico's water supply agency is required.

8. Conclusions

The Basin of Mexico Hydrogeological Database has been developed as a first step towards achieving an Integrated Water Resources Management approach in the Basin of Mexico as it provides spatial data on soils, land cover, geology and climatological variables which can be used in regional hydrogeological studies without further processing. As exemplified in this work, the BMHDB can be accessed and analyzed with the use of open source software, freely available from the Internet and completely cross-platform.

The GIS visualization capabilities provide a means to verify the spatial location of point data and to communicate the result of future water management decisions obtained through modeling, however, a GIS is not required to analyze and visualize the data stored in the RDBMS as exemplified in this work. Although this database was developed for the Basin of Mexico, this work provides a general framework on the development of spatial databases for hydrogeological modeling applicable to any watershed. The way in which the BMHDB is structured facilitates its application in both the development and calibration of distributed hydrological and groundwater models. The structure of the database allows it to be easily maintained and kept up-to-date allowing the inclusion of additional data as they become available.

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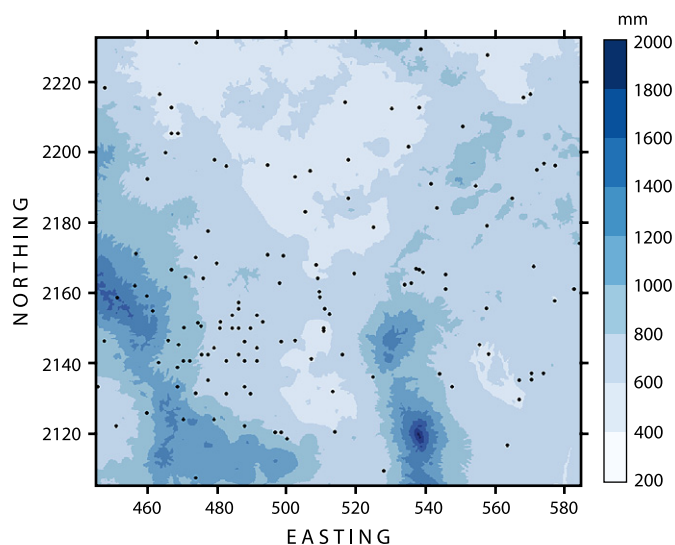


Fig. 9. Spatial distribution of rainfall for 1979; black dots represent the climatological stations used to undertake the interpolation. Coordinates are in kilometers, UTM-14.

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